

StyleGAN2

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Need for StyleGAN2



Figure 1. Instance normalization causes water droplet -like artifacts in StyleGAN images. These are not always obvious in the generated images, but if we look at the activations inside the generator network, the problem is always there, in all feature maps starting from the 64x64 resolution. It is a systemic problem that plagues all StyleGAN images.

Characteristic artifacts were observed in images generated by StyleGAN

Types of Artifacts:

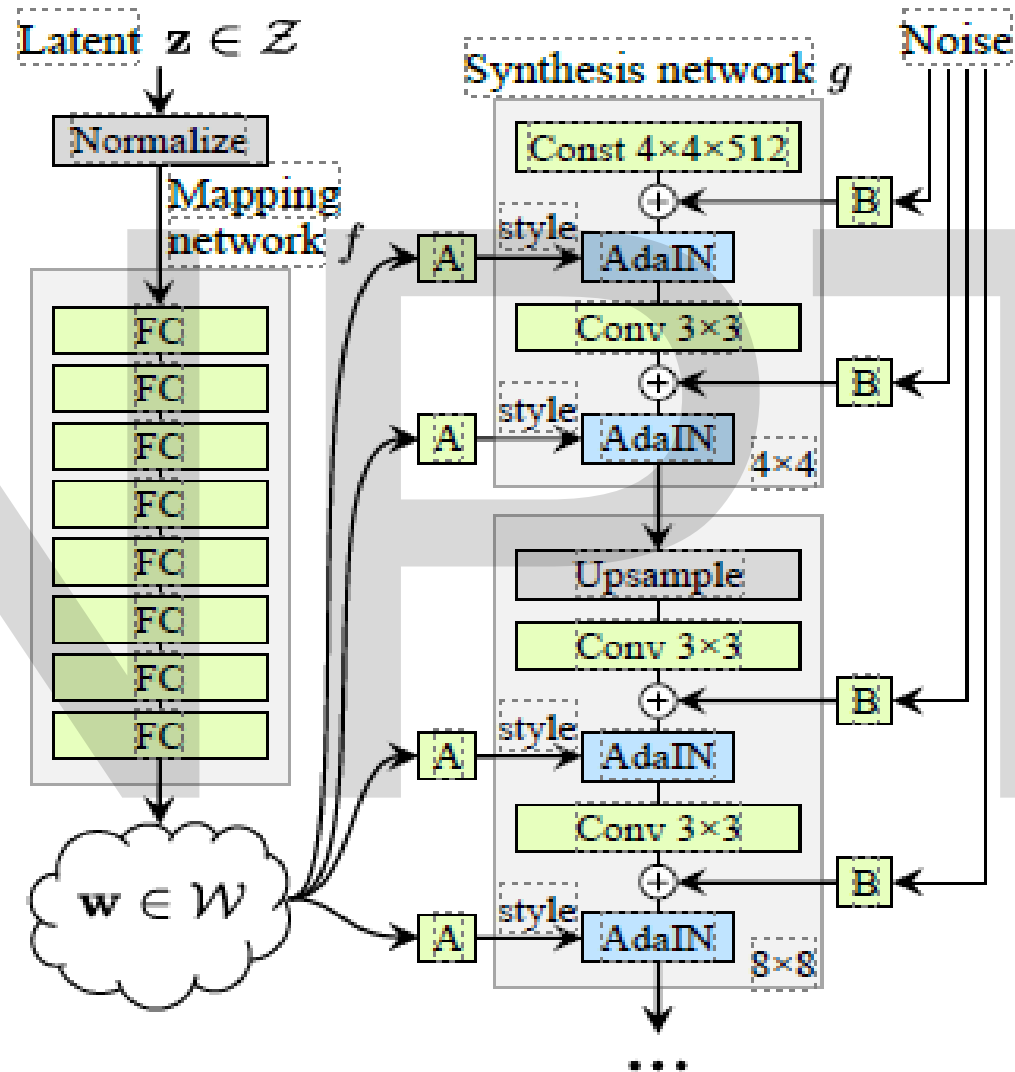
- ✓ Water-Droplet like Artifact.
- ✓ Phase Artifact (blob/phase coherence artifact)

StyleGAN2 was developed to remove artifacts created by StyleGAN while preserving image quality and controllability.

Ref:

Tero Karras, Samuli Laine, Miika Aittala, Janne Hellsten, Jaakko Lehtinen, Timo Aila “Analyzing and Improving the Image Quality of StyleGAN” (arXiv:1912.04958 [cs.CV])

Quick Review on StyleGAN



The artifact begins to appear around 64×64 resolution and becomes progressively stronger at higher resolutions.

(b) Style-based generator

Image Ref:

Tero Karras, Samuli Laine, Timo Aila, "A Style-Based Generator Architecture for Generative Adversarial Networks"

Origin of Water-Droplet-like Artifact

- ✓ AdaIN injects style at each layer by changing the mean and variance of feature map.
- ✓ But this step interfered with signal **statistics**

Mean and variance of each channel of the feature map.
(They carry semantic information.)

Before AdaIN:

- Feature mean = meaningful
- Feature variance = meaningful

AdaIN first normalizes the feature maps and then applies style-dependent scaling and shifting.

$$\text{AdaIN}(x_i, y) = y_{s,i} \frac{x_i - \mu(x_i)}{\sigma(x_i)} + y_{b,i}$$

Origin of Water-Droplet-like Artifact

The generator learns to exploit normalization statistics.

- The generator creates one extremely strong, localized spike in an image (a single pixel or tiny region with a very high value), that spike dominates the statistics that Instance Normalization computes.
- The mean and variance are now almost entirely determined by that one spike.
- **This spike manifests visually as a small blob or droplet artifact in generated images.**

Analogy

- ✓ Imagine a class where one student scores 10,000 points while everyone else scores 0–100.
- ✓ The extreme outlier "controls" the scale for everyone.

Solution to Water-Droplet-like Artifact

StyleGAN2 removed AdaIN normalization and introduced **Weight Modulation + Demodulation**.

Core Idea

Instead of modifying feature maps directly, the style modifies the convolution weights.

Weight Modulation

Weights are scaled using style values

$$\mathbf{w}'_{ijk} = s_i \mathbf{w}_{ijk}$$

Where:

s_i = style scaling factor

w_{ijk} = convolution weights

i = input channel

j = output channel

k = the location of a weight inside the convolution filter.

Weight Demodulation

- After modulation some channels may become extremely large.
- That causes instability.

Weight Demodulation

Normalizes the magnitude of the weights

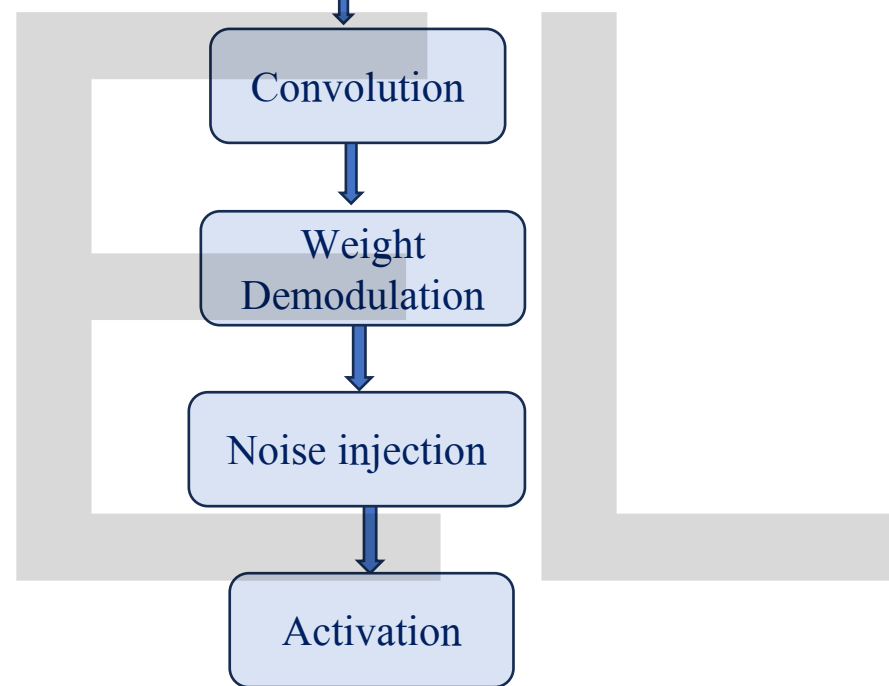
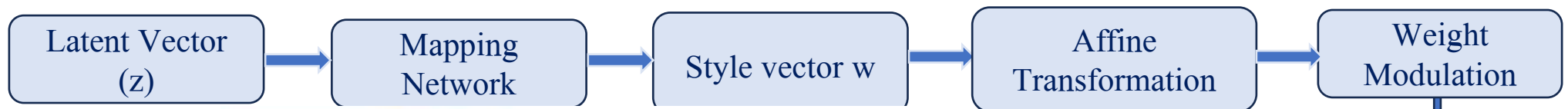
$$w_{ijk}'' = \frac{w_{ijk}'}{\sqrt{\sum_{i,k} (w_{ijk}')^2 + \epsilon}}$$

Annotations:

- w_{ijk}' : Modulated weight
- ϵ : Very small number, added to avoid division by zero.
- Squared to avoid the cancellation of positive and negative weights.
- Add over: all input channels i and all kernel positions k
- Square root restores normalized energy back to original scale

Normalizes one output channel at a time

Weight Modulation Demodulation Pipeline



Phase Artifact (blob)

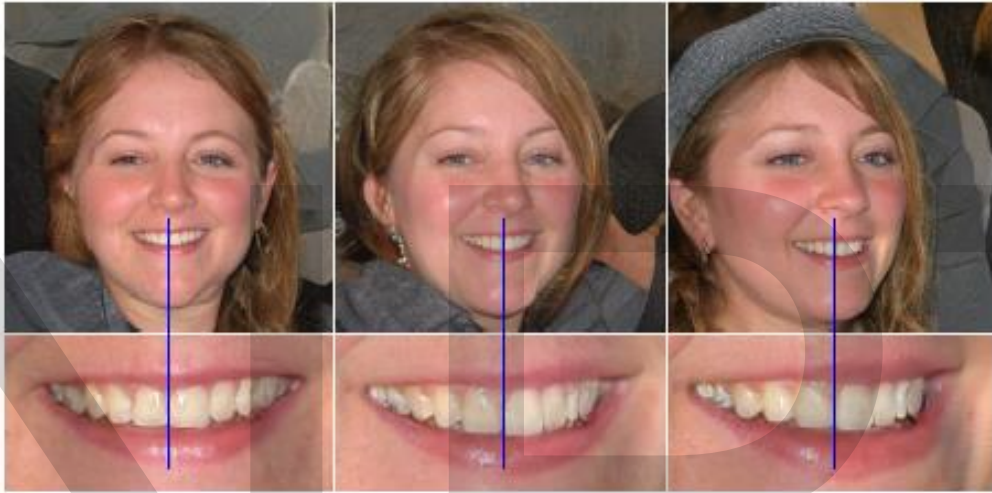


Figure 6. Progressive growing leads to “phase” artifacts. In this example the teeth do not follow the pose but stay aligned to the camera, as indicated by the blue line.

Why is it called Phase Artifact?

In signal processing, “phase” refers to spatial alignment or positional consistency of patterns.

- ✓ When the face rotates, the teeth should rotate and move with the face.
- ✓ But instead, they remain aligned to the camera and then suddenly jump.

Origin of Phase Artifact

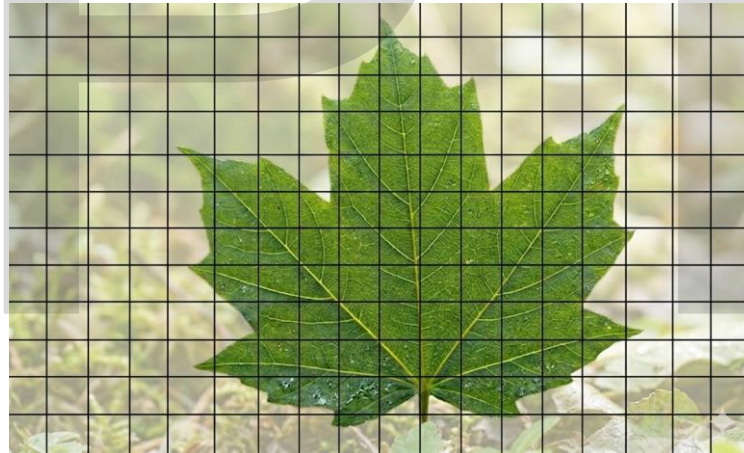
Progressive Growing in Training phase

In original StyleGAN:

- ✓ Training begins at very low resolution (4 x 4)

During training:

- ✓ Each intermediate resolution temporarily acts as final output.
- ✓ Layers are forced to generate fine details too early beyond their spatial resolution capability.



Low-frequency region

- ✓ Neighboring pixels are similar.
- ✓ Changes are small and gradual
- ✓ The image is smooth.

High-frequency region

- ✓ Neighboring pixels change sharply.
- ✓ Large jumps occur between adjacent pixels.
- ✓ Image contains sharp edges, fine texture and detailed patterns

Progressive Growing Causes High Frequency Problem

Network is training at 64×64

This layer must already create realistic images

Features become position-dependent

64×64 does not have enough spatial precision.
one pixel covers a large image region.

Problem with progressive growing

- ✓ High-frequency features are very sensitive to location.
- ✓ Small positional errors cause:
 - distorted edges
 - flickering textures
 - inconsistent spatial structures

Pixel 0	Pixel 1
0	1

Edge

After upsampling

Pixel 0	Pixel 1	Pixel 2	Pixel 3
0	0.33	0.66	1

Solution to Phase Artifact

Removed progressive growing completely

- Kept the architecture fixed
- Used Skip Connections in the Generator
- Residual Connections in the Discriminator

Training Phase

Original StyleGAN

- Stage 1: 4 x4 only
- Stage 2: New 8×8 block inserted
- Stage 3: New 16×16 block inserted
-

Architecture changes
continuously.

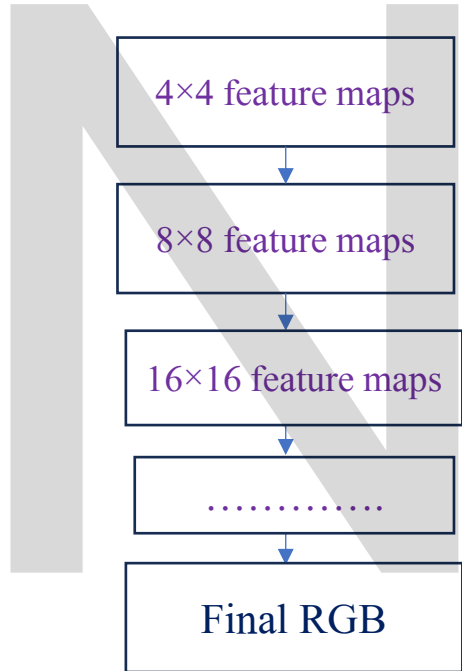
StyleGAN2

- From the first training iteration:
 $4 \times 4 \rightarrow 8 \times 8 \rightarrow 16 \times 16 \rightarrow 32 \times 32 \rightarrow \dots \rightarrow 1024 \times 1024$
- Entire network is fixed.
- No new layers are inserted later.

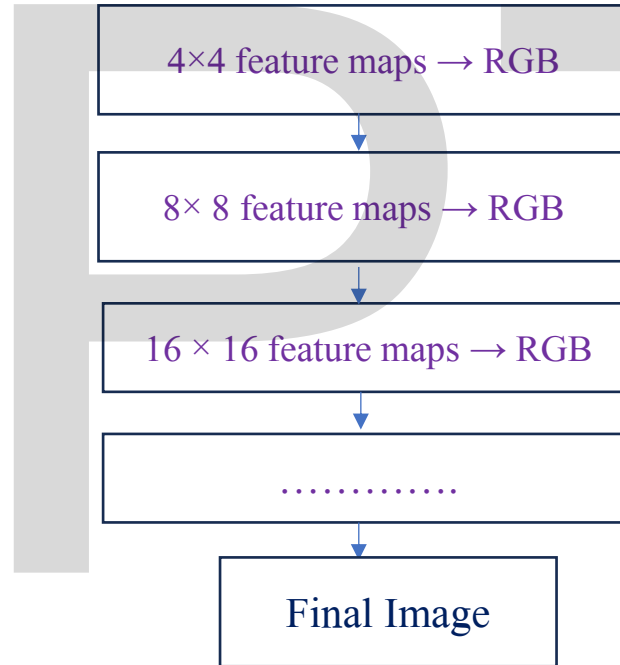
Architecture is fixed

Skip Connections in the Generator

Original StyleGAN Generator
(without Skip Connection)



StyleGAN 2
(with skip connection)



Without skip connections:

- ✓ Early layers must wait through many layers to affect output.
- ✓ Spatial shifts gets accumulated over the upsampling layers

With skip connections:

- ✓ early layers directly influence the final image.
- ✓ Multi-scale RGB outputs are progressively combined.

- ✓ Better gradient flow
- ✓ Improved training stability

Residual Connections in the Discriminator

Core Idea

Phase artifacts are basically:

- small spatial misalignments
- texture drifting
- features tied to pixel phases.

- A stronger and spatially stable discriminator forces the generator to produce properly aligned image features.

- After removing progressive growing, StyleGAN2 used a full-resolution discriminator from the start.
- This makes the discriminator deep

In original styleGAN



4 x 4



8 x 8



16 x 16



32 x 32



.....



.....



.....



1024 x 1024

✓ The discriminator initially handled only small images.

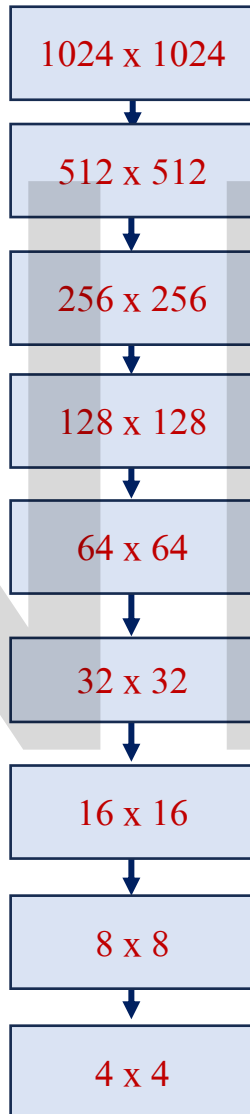
✓ Required a shallow network

When progressive growing is removed, the discriminator initially only has to process large image (1024 x 1024).

✓ The discriminator handles larger images

✓ Required a deeper network

Processing a high resolution image require many downsampling stages



Each stage contains convolution layers making the network deep from the beginning.

Problems with deep network

- ✓ Gradients may vanish.
- ✓ Training becomes unstable.
- ✓ Optimization becomes difficult

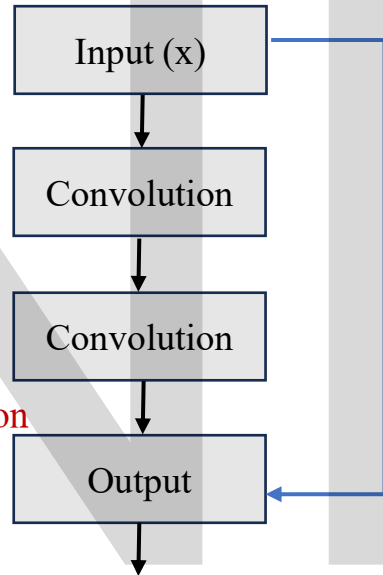
Residual Connection in Discriminator

Residual Connection

- ✓ If original input already contains useful information. The network should not destroy it.

- ✓ Residual connection ensures:
Original information + New learned refinements

$F(x)$
Learned
transformation



$$y = F(x) + x$$

Without residual paths:

Repeated convolutions may distort feature locations as original alignment information weakens.

NPTEL

Summary

Next Session

Diffusion Models